

問題 以下の英文を読んで、問(1)は 1 枚目、問(2)(3)は 2 枚目に回答しなさい。

Problem Read the following English writing, and then answer question (1) and questions (2) (3) in separate answer sheets, respectively.

①The emerging discipline of tissue engineering has the grand aim of understanding the principles of tissue growth, and applying this to produce functional replacement tissue for clinical use. There have been several [1] successes. A recent example is the work of Patrick Warnke, Hendrick Terheyden and co-workers, in which a section of replacement jaw was generated inside a sculpted titanium mesh cage by coaxing appropriate stem cells (those elusive precursor cells that give rise to specialized body tissues) to form [2]. But although successes such as this show that, in concept, tissue engineering is possible, routine implementation of such strategies remains some time off.

②In fact, this is no big surprise, as such a broad implementation requires a much better understanding of the principles of tissue formation than we currently possess, from the fundamentals of stem-cell biology to the physics and biomechanics of pattern formation. To complicate (or perhaps, enrich) matters further, this mandate also calls on the specialist expertise of scientists from a wide variety of disciplines — such as cell and molecular biologists, clinicians and materials scientists — each of whom sees the various problems involved from the perspective of their own discipline. Practical integration of these seemingly disparate strands of understanding is proving to be a rich source of scientific challenge and opportunity. In particular, [3] between biologists and mathematicians are now providing alternative and often innovative ways of thinking about tissue regeneration.

Reproducing functional tissue *ex vivo* requires an understanding not only of the behaviour of individual cells, but [4] of how global form and function arise from local cellular interactions. ③By looking at evolving tissue as a complex biological system, mathematical models can provide just such a holistic understanding. The use of agent-based models to interpret stem-cell systems is beginning to show promise in offering new ways of thinking about tissue evolution.

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注 tissue engineering 再生医療, discipline 分野, replacement tissue 代替細胞組織, stem cell 幹細胞(様々な細胞に変化する特殊な細胞), precursor cell 前駆細胞, titanium チタン製, jaw 顎, coaxing ～にうまく誘導する, routine implementation 確立された手順で実施すること, remains some time off 手付かずの状態のまま, molecular biologists 分子生物学者, seemingly disparate strand 一見すると異なる要素, behaviour 挙動・特性, *ex vivo* 体外で

問（１）下線部①～③を和訳せよ。

Question (1) Translate the underlined parts ① , ② and ③ into Japanese.

問（２） ～ に当てはまる単語を以下から番号で選びなさい。

Question (2) Choose the most suitable word for blanks ～ from the list given below and answer with the number.

- (1) also , (2) bone, (3) collaborations, (4) incorporated
(5) remarkable, (6) lattice, (7) program, (8) perspective

問（３）組織形成の原理をより深く理解するためには何が必要だと言っているかを，簡潔に日本語か英語で記述しなさい。

Question (3) Briefly explain what is necessary to understand better the principles of tissue formation , in Japanese or English.